Worksheet 4.1 — Sampling and Presenting Data

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

Good sampling avoids bias; grouped data needs a midpoint estimate for its mean.

- Stratified sampling: sample each subgroup proportionally to its size.
- Estimated mean from grouped data: $\bar{x} \approx \dfrac{\sum fm}{\sum f}$, using class midpoints m.

Worked Examples

Example 1 — Stratified sample sizes

A population of 1000 has 350 in group P, 650 in group Q. Find stratified sample sizes for a total sample of 40.

Solution. Fraction $= 40/1000 = 1/25$. P: $350/25 = 14$. Q: $650/25 = 26$.

Conclusion: 14 from P, 26 from Q.

Example 2 — Bias critique

A survey on gym habits only interviews people leaving a gym at 6am. Identify the bias.

Solution. This is convenience sampling of a specific subgroup (early-morning gym-goers), likely the most dedicated exercisers.

Conclusion: over-represents highly motivated exercisers; underestimates how rarely the average person exercises.

Example 3 — Grouped mean

Estimate the mean from: 0–10 (freq 5), 10–20 (freq 8), 20–30 (freq 7).

Solution. Midpoints $5, 15, 25$. $\sum fm = 25 + 120 + 175 = 320$; $\sum f = 20$.

Conclusion: $\bar{x} = 16$.

Example 4 — Modal class

State the modal class for the data in the previous example.

Solution. Highest frequency (8) is in the 10–20 class.

Conclusion: modal class 10–20.

Example 5 — Systematic sampling

A quality inspector checks every 20th item off a production line of 2000 items. State the sampling method and the sample size.

Solution. This is systematic sampling (constant interval $k = 20$).

Conclusion: sample size $= 2000/20 = 100$ items.

Example 6 — Estimating the median from grouped data

Estimate the median class for the grouped data 0–10(4), 10–20(9), 20–30(7).

Solution. Total frequency = 20; the median (10th/11th value) falls where the cumulative frequency first reaches 10: after 4 (in 0–10) plus 9 (in 10–20) = $13 \geq 10$.

Conclusion: median class is 10–20.

Example 7 — Comparing two grouped means

Two classes' test scores are grouped as: Class A — 60–70(3), 70–80(9), 80–90(3); Class B — 60–70(6), 70–80(6), 80–90(3). Compare their estimated means.

Solution. Class A: $\bar{x} = \dfrac{3(65) + 9(75) + 3(85)}{15} = \dfrac{195 + 675 + 255}{15} = \dfrac{1125}{15} = 75.$

Class B: $\bar{x} = \dfrac{6(65) + 6(75) + 3(85)}{15} = \dfrac{390 + 450 + 255}{15} = \dfrac{1095}{15} = 73.$

Conclusion: Class A's estimated mean (75) is slightly higher than Class B's (73).

Example 8 — Application — stratified sample by three groups

A survey of 900 residents (500 downtown, 250 suburban, 150 rural) needs a stratified sample of 90. Find each group's sample size.

Solution. Fraction = $90/900 = 1/10$.

Conclusion: downtown 50, suburban 25, rural 15.

Example 9 — Identifying a sampling frame issue

A phone survey uses only listed landline numbers to estimate opinions of all adults. Explain a limitation of this sampling frame.

Solution. Many adults, especially younger ones, don't have listed landlines — they're mobile-only or unlisted.

Conclusion: the sampling frame excludes a large, systematically different portion of the population, biasing the results.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

A population of 600 has 220 in group A, 380 in group B; find stratified sample sizes for a sample of 30.

Question 2

Estimate the mean from: 0–4 (freq 6), 4–8 (freq 9), 8–12 (freq 5).

Question 3

State the modal class for the data in the previous question.

Question 4

A researcher only surveys people who respond to an online ad; explain a likely bias.

Question 5 — Challenge

Estimate the mean from: 10–20 (freq 3), 20–30 (freq 8), 30–40 (freq 12), 40–50 (freq 7).

Question 6

A school of 1200 (700 juniors, 500 seniors) wants a stratified sample of 60; find sample sizes for each group.

Question 7

Explain why a census (surveying an entire population) is not always more useful than a well-designed sample.

Question 8

An inspector checks every 15th item from a batch of 1500; state the method and sample size.

Question 9

Estimate the median class for: 0–5(2), 5–10(6), 10–15(8), 15–20(4).

Question 10

A survey of 800 residents (450 urban, 200 suburban, 150 rural) needs a stratified sample of 80; find each group's sample size.

Question 11 — Challenge

A university surveys only students in the library on a Friday night to estimate typical study habits; explain the bias.

Question 12 — Challenge

Two classes' scores are grouped: Class C — 50–60(4),60–70(10),70–80(6); Class D — 50–60(8),60–70(6),70–80(6). Compare their estimated means.

Question 13 — Challenge

A pollster only calls numbers from a decade-old phone directory; explain a limitation of this sampling frame.

Question Recap & Answer Key

Question Recap

1. A population of 600 has 220 in group A, 380 in group B; find stratified sample sizes for a sample of 30.
2. Estimate the mean from: 0–4 (freq 6), 4–8 (freq 9), 8–12 (freq 5).
3. State the modal class for the data in the previous question.
4. A researcher only surveys people who respond to an online ad; explain a likely bias.
5. Estimate the mean from: 10–20 (freq 3), 20–30 (freq 8), 30–40 (freq 12), 40–50 (freq 7).
6. A school of 1200 (700 juniors, 500 seniors) wants a stratified sample of 60; find sample sizes for each group.
7. Explain why a census (surveying an entire population) is not always more useful than a well-designed sample.
8. An inspector checks every 15th item from a batch of 1500; state the method and sample size.
9. Estimate the median class for: 0–5(2), 5–10(6), 10–15(8), 15–20(4).
10. A survey of 800 residents (450 urban, 200 suburban, 150 rural) needs a stratified sample of 80; find each group's sample size.
11. A university surveys only students in the library on a Friday night to estimate typical study habits; explain the bias.
12. Two classes' scores are grouped: Class C — 50–60(4),60–70(10),70–80(6); Class D — 50–60(8),60–70(6),70–80(6). Compare their estimated means.
13. A pollster only calls numbers from a decade-old phone directory; explain a limitation of this sampling frame.

✓ Answer Key

1. fraction $= 1/20$: A: 11, B: 19
2. midpoints $2, 6, 10$; $\sum fm = 12 + 54 + 50 = 116$, $\sum f = 20$: $\bar{x} = 5.8$
3. 4–8 (freq 9)
4. over-represents people active online / interested enough to respond, under-represents everyone else
5. midpoints $15, 25, 35, 45$; $\sum fm = 45 + 200 + 420 + 315 = 980$, $\sum f = 30$: $\bar{x} \approx 32.7$
6. fraction $= 1/20$: juniors 35, seniors 25
7. a census can be slower, costlier, and harder to keep accurate/consistent across every member than a well-designed representative sample
8. systematic sampling; sample size $= 1500/15 = 100$
9. cumulative: $2, 8, 16, 20$; the median position (10th value) falls once cumulative reaches ≥ 10, in the 10–15 class
10. fraction $= 80/800 = 1/10$: urban 45, suburban 20, rural 15
11. over-represents heavy studiers who use the library late on a Friday, under-represents typical/lighter studiers
12. Class C: $\bar{x} = \frac{4(55)+10(65)+6(75)}{20} = \frac{220+650+450}{20} = \frac{1320}{20} = 66$; Class D: $\bar{x} = \frac{8(55)+6(65)+6(75)}{20} = \frac{440+390+450}{20} = \frac{1280}{20} = 64$; Class C's mean is slightly higher
13. the directory is outdated — misses newer residents, movers, and increasingly excludes people without landlines, skewing who gets reached